

Workshop 2.3: Machine Learning vs. Deep Learning

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In the field of Artificial Intelligence, there are many terms that have similar, yet distinct, definitions, and understanding the differences between them is paramount to having successful conversations with colleagues or writing effectively about a specific topic. Two of these terms are machine learning (often shortened to ML) and deep learning (often shortened to DL). Both are approaches to getting a computer to perform a task, but they differ in how much of the work the human does versus how much the computer figures out on its own.

To see what makes them distinct, it helps to start with what came before them: traditional programming. Imagine the solution guide that comes with a Rubik's Cube. It gives you a fixed starting point (find the white center) and then a branching series of "if the cube looks like this, do that" instructions that always lead to a solved cube. A programmer can translate that same guide into code, and a machine with motors and cameras will solve the cube every time. However, in this scenario the machine isn't "learning" anything, it's merely following directions. If you placed a different sized cube instead of the standard 3x3, or even changed the colors of the squares, the computer would be unable to complete the task.

Machine learning departs from this model in a fundamental way. Instead of a human writing out the rules, the computer learns the rules from data (Fireship, 2022). A great example of machine learning in action is spam filtering. In the traditional approach, a person would write a list of banned words and sender addresses, the cube-guide method applied to email. A machine learning spam filter works differently: it is shown thousands of emails already labeled "spam" or "not spam," and it learns on its own which patterns tend to separate the two. A human still plays a role, choosing which features the model should pay attention to (word frequency, sender domain, how many recipients an email went to) and supplying the labeled examples to learn from. But the human does not write the decision rule itself. The model derives that from the data. This is the key shift: the human provides the examples and the relevant signals, and the algorithm produces the model. There are tradeoffs, however. Because the model's behavior comes from patterns it learned rather than rules a human wrote, there is no line of code to open and edit. If you wanted it to stop flagging emails about a particular topic, you couldn't simply rewrite a rule; you would have to correct the labels on those examples in the training data and retrain the model so it learns the new pattern.

Conversely, deep learning is a subcategory of machine learning, and uses a much more sophisticated method during its training process. "Deep learning models identify, classify, and analyze structured data, images, text, and sound" (Dilmegani & Sezer, 2026). So, deep learning is trained on data like all machine learning, but uses a neural net with multiple layers to achieve an even greater comprehension of the data. Once trained, it can process complex, unstructured input (like live camera feeds) and react to changing situations in real time, something simpler machine learning models struggle with. The learning itself happens beforehand during training; what the deployed model does in the moment is apply what it already learned. This allows it to be used in applications that are much more difficult and require much more data; such as autonomous driving. Autonomous driving is the process of using an onboard computer within a car, along with an array of cameras and other sensors, to drive the car either with partial human involvement or completely by itself. This is possible in large part due to computer vision: which allows a computer to understand its environment through capturing images, processing them, and then classifying the contents (Dilmegani & Sezer, 2026). When an autonomous vehicle is in

motion, the input from the cameras is constantly monitoring for an array of different road hazards, other vehicle indicators, and traffic directions in order to react to the ever changing conditions.

It may be easy to assume that since deep learning deals with more data, in real time, than other types of machine learning, it is the superior learning method and should be used in all circumstances. However, this is not the case for a variety of reasons. For the spam filter example used above, this relatively simple model can be trained and deployed to thousands of clients, with little to no impact on computing resources. When an update or security measure is necessary, the update can be applied to a single host system for testing purposes before being rolled out en masse. This allows for issues with the model to appear before it is in use, which increases the reliability of the product. If a deep learning model were used as a spam filter, it would demand vastly more computing resources because the model itself is large: many layers and millions of parameters mean every email it evaluates requires far more computation than a simple algorithm needs. Training such a model in the first place also requires enormous amounts of labeled data and significant processing power, an investment that buys little benefit for a task a lightweight model already handles well. Updating its behavior later means retraining and revalidating that large model, which is slower and more costly than adjusting a simple one.

Likewise, machine learning could not be used for something like autonomous driving without the aid of computer vision, which requires a neural net with multiple layers. Since autonomous vehicles must operate at speed, the real time inferencing of data is paramount to ensuring the vehicle and its occupants arrive safely. A human developer could identify thousands of potential road hazards, but because of the non-uniformity of the real world it would be impossible to identify every hazard possible. Therefore, the computer would not have the information it needed to react appropriately when it did reach an unnamed hazard.

For these reasons, neither model is “superior” to the other, they both have valid and legitimate use cases. Classical machine learning is best when you have a large amount of labeled and structured data and real time inferencing is not necessary. It is also useful when computing resources are constrained or when it needs to be running alongside other applications inside of a network or computer. Deep learning’s strengths are its ability to process large and unstructured data that is difficult or impossible for humans to accurately label, and then use that data to inference in real time. Overall, the two models are different solutions to different problems, and identifying which is the better fit will factor into the model’s success.

References

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